

Book-of-Life Crime Prediction with NLSY79 Child and Young Adult Data: Workflow, Target Construction, Feature Design, Pilot Results, and Limitations

Seminar Project Working Memo

May 29, 2026

1 Purpose of this memo

This document summarizes the workflow developed for the seminar project on predicting later delinquency and justice contact using NLSY79 Child and Young Adult data. It is meant as a long-form project note that records what was done, why particular choices were made, what the current files contain, and what the main methodological caveats are.

The project began with an exploratory analysis of the original NLSY79 cohort crime items and gradually shifted toward the NLSY79 Children and Young Adults data. The reason for this shift is substantive and methodological: the child and young adult data contain repeated observations over a longer time span, which makes them more suitable for a Book-of-Life (BoL) representation. Instead of representing a person as a single row of tabular variables, the BoL approach converts temporally ordered life-course information into structured text. That text can then be read directly by an LLM or embedded and used in downstream prediction models.

The current direction is to keep the broad persistent delinquency/contact target as the main analysis and add a second, closely related reoffense framing. The reoffense version conditions on having a first observed direct delinquency/contact event and asks whether a later second event occurs. This lets us compare two interpretations of the same event history: predicting persistence in the full observed target cohort versus predicting reoffense among people who have already had a first event.

2 Research questions

The current project is organized around three main research questions:

- RQ1. How well can LLM-based Book-of-Life representations predict later delinquency or justice contact compared with tabular baselines?
- RQ2. How do fairness and error-rate patterns differ across sex and race/ethnicity groups for LLM-based predictions and tabular baselines?
- RQ3. Which types of life-course information appear most predictive: demographic background, family context, school attachment, behavior problems, substance use, prior delinquency/contact, or other domains?

RQ4. Does the interpretation change if the first event is treated as an observed index event and the target is reframed as later reoffense?

The project is exploratory. The goal is not to build an operational risk tool, but to study whether rich text representations of survey histories add predictive value and how such models behave in a sensitive domain.

3 Data sources and current processed data

3.1 Initial NLSY79 parent/cohort analysis

The first phase used the original NLSY79 respondent data. That data contained basic controls such as sample ID, sex, race/ethnicity, age, and 1980 self-reported crime items. We constructed targets such as:

- any delinquency,
- any violent delinquency,
- any property delinquency,
- drug selling,
- a severity scale ranging from no crime to more severe offending.

This phase was useful for learning the codebook and testing baseline workflows, including LASSO logistic regression, grouped feature importance, base-rate comparisons by sex and race, and fairness diagnostics. However, the original parent/cohort extract was limited for the BoL approach because many predictors were concentrated around 1979 and the crime outcome was measured in 1980. This meant that the available life history was relatively short.

3.2 NLSY79 Children and Young Adults data

The later phase used a combined NLSY79 Children and Young Adults dataset that was created from 18 downloaded folders and their separate codebooks. The combined files were placed in the shared-team bundle:

- `output/share_with_team/nlsy79_child_youngadult_combined_bundle/`
- `nlsy79_child_youngadult_selected_crime_features.csv`
- `nlsy79_child_youngadult_selected_crime_features_codebook.csv`
- `nlsy79_child_youngadult_codebook_unique.csv`
- `nlsy79_child_youngadult_codebook_long.csv`

The selected child/young-adult crime-feature dataset contains:

- 4,739 unique children,
- 2,022 unique mothers,

- child birth years from 1971 to 2014,
- median child birth year 1987,
- 2,453 male children and 2,286 female children,
- 436 Hispanic children, 686 Black children, and 3,617 non-Black/non-Hispanic children under the NLSY race/ethnicity coding.

The relevant variable years in the selected subset range from 1986 to 2020. The longitudinal structure is therefore much more appropriate for BoL-style prediction than the initial 1979/1980 parent-cohort setup.

4 Codebook structure

Two versions of the combined codebook were created:

Long codebook

Contains one row per source occurrence. This version is useful for tracing where a variable came from and for resolving duplicates across source folders.

Unique codebook

Contains one row per unique variable. This is easier for team sharing and general modeling because it avoids duplicate metadata rows.

For the team bundle, source-folder references such as `c_nls1` were removed from the shared codebook because they were useful during construction but confusing for end users.

Topic labels were also added to the child/young-adult codebook, inspired by the NLS Investigator topic structure. Examples include Education, Training and Achievement Scores; Employment; Household, Geography and Demographics; Health; Attitudes, Expectations and Non-cognitive Tests; and Crime and Substance Use. The topic labels are approximate metadata aids. They are not perfect semantic matches to the original parent NLSY79 R-reference variables.

5 Identifiers and merging with mother data

The child data contain both child-level and mother-level identifiers. The key practical identifiers used in the project are:

- C0000100: child identifier,
- C0000200: mother identifier.

The child identifier is unique at the child level in the processed dataset. The mother identifier is shared by siblings, which makes it possible to merge child records with mother-level information from the original NLSY79 mother data. A mother-merged version was created for sharing. The merge is conceptually possible because each child is linked to an NLSY79 female respondent, but one must remember that several children can map to the same mother.

6 Why target construction was revised

6.1 The initial 1994–1998 repeated justice-contact target

An early child/young-adult pilot target used later justice-contact items from 1994, 1996, and 1998. The target was conceptually clean because it focused on probation or correctional/jail/reform contact, but it had an important limitation: using only these years excluded many younger children who were not yet old enough or not yet observed in the relevant modules. It also did not use crime-related variables available after 1998.

This produced a pilot that was useful technically but less satisfactory for the final project because it did not leverage the full 1971–2014 birth cohort range or the later survey waves through 2020.

6.2 The need for person-specific temporal cutoffs

The central methodological issue is leakage. If a person has the target event in 2010, predictors from 2012 or 2020 cannot be used. But if another person has no target event until 2020, predictors through 2018 may be valid. Therefore, a single global cutoff year is too restrictive. It either removes too much information for later cases or leaks future information for earlier cases.

The revised BoL workflow uses a person-specific cutoff:

- for positive persistent cases, the book is cut off before the second positive target year;
- for negative cases, the book is cut off before the last observed target year.

This design allows the representation to use as much pre-outcome information as possible without using information after the event that defines the target. It also supports the reoffense framing: the first positive event can be treated as observed history, while the second positive event remains the outcome cutoff for positive reoffense cases.

7 Current target options

7.1 Option A: later persistent delinquency/contact

The current main target remains:

$$Y_i = \text{later_persistent_delinquency_contact}_i.$$

It equals 1 if a child or young adult has direct delinquency/contact indicators in at least two different survey years from 2000 to 2020. It equals 0 if the child has target information observed but does not meet that repeated-event definition.

The target intentionally combines two ideas:

1. self-reported delinquent/aggressive behavior, such as fighting, hurting someone badly, or damaging property;
2. justice-system contact, such as probation or being sentenced to corrections, jail, or reform.

The aim is to avoid a target that is too rare, while still distinguishing one-time or minor incidents from more persistent patterns.

7.2 Option B: reoffense after a first event

An additional target option reframes the task as reoffense. In this version, the analytic cohort is restricted to children and young adults with a first observed direct delinquency/contact event:

$$\text{later_any_delinquency_contact}_i = 1.$$

Within that cohort, the outcome is:

$$Y_i = \text{later_reoffend_after_first_delinquency_contact}_i.$$

It equals 1 if a second positive direct delinquency/contact year is observed after the first event year. It equals 0 if the person has a first event but no later second positive event during observed follow-up. People with no first event are out of cohort for this target.

This target is numerically equivalent to the persistent target within the first-event cohort, but the substantive question is different. Option A asks who develops repeated delinquency/contact in the full observed cohort. Option B asks, conditional on a first event, who later reoffends. This makes the first event an index/history feature rather than the event being predicted from baseline.

7.3 Target items

The broad target uses direct later delinquency/contact items from 2000–2020. Examples of item families are:

- physical fight at school or work,
- hurt someone badly enough to need attention,
- damaged school or other property on purpose,
- ever or since last interview been on probation,
- ever or since last interview been sentenced to corrections, jail, or reform.

Indirect risk-only items, such as seeing another student carry a weapon, were not used as target-defining items. They may be useful as predictors when they occur before the cutoff, but they are not the outcome.

7.4 Target sample and base rates

The full selected child dataset contains 4,739 children. For the broad target:

- 4,122 children have enough target information to classify the outcome,
- 617 children are missing the target because no relevant target year is observed,
- 3,064 children have any later delinquency/contact,
- 2,109 children have persistent later delinquency/contact.

Thus:

Target	Eligible N	Missing N	Positive N	Base rate
Any later delinquency/contact	4,122	617	3,064	0.743
Persistent later delinquency/contact	4,122	617	2,109	0.512
Reoffense after first event	3,064	1,675	2,109	0.688

The persistent target is currently preferred as the main outcome because it is balanced enough for pilot prediction while being more substantively meaningful than a very broad “any event” target. The reoffense target is useful as an additional analysis because it isolates repetition after a first observed event.

7.5 Timing of positive events

For the any-event target, first positive event years were distributed as follows:

Year	2000	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Count	438	812	413	391	313	235	186	118	115	27	16

For the persistent target, the second positive event year, which defines the positive cutoff, was distributed as follows:

Year	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Count	265	443	359	336	265	156	116	68	77	24

8 Alternative target options

Several target options remain plausible. They answer slightly different substantive questions:

Any delinquency/contact

Positive if any direct later delinquency or justice-contact item is observed. This gives a high base rate and large sample, but it mixes very minor, one-time incidents with more persistent behavior.

Persistent delinquency/contact

Positive if direct delinquency/contact appears in at least two different survey years. This is the current main target. It better captures repeated risk and avoids treating one isolated event as equivalent to persistence.

Reoffense after first event

Restricted to people with a first observed direct delinquency/contact event. Positive if a later second positive event is observed. This is the current additional target option because it treats the first event as observed history rather than as the endpoint.

Justice contact only

Positive if probation or correctional/jail/reform contact occurs. This is conceptually severe and administratively meaningful, but it is rarer and is more affected by policing, sentencing, and institutional bias.

Violent/aggressive behavior

Positive if fighting or hurting someone badly is reported. This may align with a more behaviorally specific criminological construct, but it excludes property and justice-contact outcomes.

Severity scale

A multi-level outcome distinguishing no event, one minor/non-contact event, repeated non-severe events, and severe/contact events. This is richer but harder to model and explain.

Early warning target

Positive if a person who has early mild behavior later escalates to repeated or severe behavior. This is closest to a prevention-oriented classifier but requires careful definition of an index age or index wave.

For the seminar project, the persistent target is the cleanest main target. The reoffense-after-first-event version should be reported as the main alternative framing, while any-event and justice-contact-only targets can be used as further robustness checks.

9 Feature construction

9.1 Initial selected feature subset

Before the BoL expansion, we created a smaller selected-feature dataset for the team. It included:

- demographic controls,
- theoretically motivated variables suggested by the team,
- variables selected or highlighted in earlier LASSO analyses,
- crime-related and substance-use variables,
- optional mother-level variables in a separate merged dataset.

This subset was useful for interpretable tabular modeling and team sharing. However, a very small feature set underuses the strengths of BoL text representations because much of the rich longitudinal information is omitted.

9.2 Expanded BoL feature set

The current broad-persistent BoL feature index contains 864 variables. These variables span survey years 1986–2020, but each individual book includes only features observed before that person’s cutoff. Therefore, a 2020 variable can appear in the feature index but will not appear in a book for someone whose event cutoff is earlier than 2020.

An important clarification is that the current 864 count is year-specific. Each `csv_code` is treated as its own feature, even if the same question appears again in another survey year. If identical question text is collapsed across years, the feature index contains 135 question-level features. This separates the number of raw year-specific columns from the number of conceptually distinct repeated questions.

The expanded features are grouped as follows:

Feature group	Year-specific	Question-level	Difference
Behavioral/externalizing	492	18	474
Prior delinquency/contact	141	26	115
Substance/risk behavior	87	25	62
School/achievement	72	22	50
Family/home	28	16	12
Socioeconomic/work	12	3	9
Other	11	11	0
Health	8	7	1
Peer/neighborhood	7	1	6
Background	6	6	0
Total	864	135	729

The current BoL text generator still uses the year-specific feature index, which is appropriate for preserving temporal order in the books. For tabular baselines or compact summaries, the collapsed question-level representation may be a useful alternative because it avoids treating every repeated wave as a fully separate construct.

The feature index is saved at:

```
bolt_nlsy79_starter_v0-2/data/processed_child_crime_broad_persistent/  
child_crime_broad_persistent_feature_index.csv
```

9.3 Why include prior delinquency/contact variables?

Including prior delinquency/contact variables is conceptually defensible for the persistent target as long as they occur before the cutoff. The task then becomes: given earlier life-course context and earlier risk/contact information, can we predict persistent later delinquency/contact? This is close to an early warning or persistence-prediction setup.

However, this choice changes the interpretation. The model is not purely predicting later behavior from childhood family and school background. It is also allowed to use earlier behavioral signals. For a stricter prevention question, one could run a second version excluding all prior delinquency/contact items and using only background, family, school, health, peer, and non-criminal-risk variables.

10 Book-of-Life generation

10.1 Original starter limitation

The Book-of-Life starter code was useful as a template but did not fully solve the longitudinal cutoff problem. In particular, the starter was closer to a fixed-sample/fixed-text generator. For this project, that is not enough because the valid prediction window differs across individuals.

If a target is observed in 2004 for one child and in 2018 for another child, the model should not see the same maximum calendar year for both people. The correct cutoff must depend on the person’s outcome timing.

10.2 Changes made

The main changes were implemented in:


```
bolt_nlsy79_starter_v0-2/create_child.bol_broad_persistent.py
```

The script now:

1. reads the broad persistent target file;
2. computes a person-specific `cutoff_year`;
3. builds a broad domain-relevant feature index;
4. excludes constructed target variables and post-cutoff features;
5. groups variables into interpretable domains;
6. renders one structured Book-of-Life text per eligible child;
7. writes the full 4,122-book dataset and a balanced 120-case pilot sample.

The current output files are:

- `books_child_crime_broad_persistent_cutoff.json`
- `books_child_crime_broad_persistent_sample_120.json`
- `child_crime_broad_persistent_targets.csv`
- `child_crime_broad_persistent_feature_index.csv`
- `child_crime_broad_persistent_sample_ids_120.csv`

The reoffense analysis currently reuses these generated books and prediction probabilities rather than regenerating a separate reoffense-prompted BoL sample. A future version should generate a dedicated reoffense cohort and use a prompt that explicitly states that every person has already had a first event.

10.3 Example text structure

Each book begins with stable background information and then lists pre-cutoff information by year and domain. A simplified structure is:

This Book of Life describes NLSY79 child respondent [ID], linked to mother ID [ID]. Only information observed before [cutoff year] is included. Stable background information: sex, race/ethnicity cohort, birth year, birth order, and maternal age at birth. In 1992: behavioral/externalizing: behavior problem items, antisocial scores, externalizing scores. In 1994: family/home: household composition; behavioral/externalizing: updated behavior problem items; school: achievement and school context. ...

This representation keeps the information structured rather than turning it into speculative narrative. That is important because the LLM should not invent psychological explanations that are not directly observed.

10.4 A note on LLM “thinking” and interpretation

We discussed whether the LLM should be allowed to add implications such as “moving often may have produced an unstable sense of belonging.” This could be useful for qualitative interpretation, but it is risky for prediction. It can introduce unobserved assumptions, stereotypes, and race/class/gender-coded biases. Therefore, the current recommendation is:

- use neutral, structured BoL text for the main prediction task;
- optionally run a separate interpretive-summary condition as an ablation;
- never treat speculative LLM inferences as observed facts.

11 LLM prediction setup

11.1 Direct prompting

For the pilot, we used direct prompting: the model reads each Book of Life and returns JSON containing a binary prediction, a probability, and a one-sentence reason. The prediction script is:

```
bolt_nlsy79_starter_v0-2/predict_child.bol.broad.persistent.openai.py
```

The current model is `gpt-5.4-nano`, chosen because it is cheap enough for pilot experiments. The first direct-prompting runs tended to produce only negative predictions, even when probabilities varied. We therefore revised the prompt to:

- define the target explicitly;
- tell the model that the book is already cut off before the target point;
- calibrate risk examples verbally;
- require valid JSON;
- require the binary prediction to equal 1 when probability is at least 0.50.

The notebook now keeps two prompt templates: Option A for broad persistent delinquency/contact, and Option B for reoffense after a first event. The current saved 120 predictions were produced with the Option A prompt. The Option B results below are therefore a derived relabeling and evaluation of the same probabilities, not a fresh LLM run with a reoffense-specific prompt.

11.2 Threshold evaluation

Because LLM probabilities are not guaranteed to be calibrated, we evaluate thresholds after prediction. The threshold script is:

```
bolt_nlsy79_starter_v0-2/evaluate_child.bol.broad.persistent.thresholds.py
```

For the current balanced 120-case pilot, the best accuracy threshold was 0.38. This should not be interpreted as a final deployed threshold because it was selected on the same small pilot sample.

12 Pilot results

12.1 Current 120-case broad persistent pilot

The 120-case sample was balanced: 60 positive and 60 negative cases. The average generated book length was approximately 8,254 characters after the expanded feature set was added.

The current LLM pilot results are:

Threshold	Accuracy	AUC	TPR	TNR	FPR	FNR	TP/TN	FP/FN
0.38	0.858	0.895	0.867	0.850	0.150	0.133	52/51	9/8
0.34	0.850	0.895	0.900	0.800	0.200	0.100	54/48	12/6
0.35	0.850	0.895	0.867	0.833	0.167	0.133	52/50	10/8

This is a strong improvement over the earlier narrow-feature and less-calibrated prompt runs. However, the result must be treated as preliminary because:

- the sample is small;
- the sample is artificially balanced;
- the best threshold was chosen on the same data;
- there is not yet a large held-out comparison to a tabular baseline using the same target, features, and cutoff logic.

12.2 Derived reoffense evaluation on the original 120 runs

We also derived an Option B reoffense evaluation from the original 120 broad-persistent prediction rows. This keeps the model probabilities unchanged and restricts the evaluation to cases with a first observed direct delinquency/contact event. Of the original 120 rows, 85 are in this first-event cohort:

- 60 are reoffense positives, meaning a later second event is observed;
- 25 are reoffense negatives, meaning a first event is observed but no later second event is observed;
- 35 original rows have no first event and are excluded from the reoffense cohort.

The derived reoffense threshold evaluation is:

Threshold	Accuracy	AUC	TPR	TNR	FPR	FNR	TP/TN	FP/FN
0.38	0.835	0.828	0.867	0.760	0.240	0.133	52/19	6/8
0.34	0.824	0.828	0.900	0.640	0.360	0.100	54/16	9/6
0.36	0.824	0.828	0.867	0.720	0.280	0.133	52/18	7/8
0.50	0.647	0.828	0.567	0.840	0.160	0.433	34/21	4/26

This result is useful because it shows that the original predictions retain signal when the task is reframed as reoffense. However, it should be described as a derived sensitivity analysis. A cleaner next step is to regenerate a reoffense-specific sample and prompt the model with the Option B target definition from the start.

12.3 Fairness results for the 120-case pilot

At threshold 0.38, subgroup metrics were:

Attribute	Subgroup	N	Base rate	Pred.+	Accuracy	AUC	FPR/FNR
Sex	Female	58	0.448	0.517	0.897	0.932	0.156/0.038
Sex	Male	62	0.548	0.500	0.823	0.867	0.143/0.206
Race	Black	22	0.500	0.500	0.818	0.913	0.182/0.182
Race	Hispanic	13	0.308	0.462	0.846	0.875	0.222/0.000
Race	Non-Black/non-Hispanic	85	0.529	0.518	0.871	0.892	0.125/0.133

The main subgroup caveat is that the Hispanic subgroup has only 13 cases in the balanced pilot. Fairness estimates for such small subgroups can move substantially if only a few predictions change.

12.4 Fairness gaps

The largest sex gaps were in true positive and false negative rates:

- sex predicted-positive-rate gap: 0.017,
- sex accuracy gap: 0.074,
- sex AUC gap: 0.065,
- sex TPR/FNR gap: 0.167.

Race gaps were:

- race predicted-positive-rate gap: 0.056,
- race accuracy gap: 0.052,
- race AUC gap: 0.038,
- race TPR/FNR gap: 0.182,
- race PPV gap: 0.220.

The PPV gap is partly affected by different subgroup base rates and small subgroup sizes.

13 Why BoL representations might help

BoL text representations may have a predictive advantage over tabular baselines for several reasons:

1. They preserve variable labels and semantic context. A tabular model sees a code such as C0566600; the LLM sees that the item concerns being disobedient at school.
2. They preserve temporal order. The model can see whether problems appear early, persist, or emerge later.

3. They allow related variables to be interpreted jointly. For example, school problems, externalizing behavior, and prior substance use may form a recognizable pattern.
4. They reduce the need for hand-engineered interactions, because the text can describe combinations of events directly.
5. They can incorporate broad language-model knowledge about what variable descriptions mean.

The last point is both useful and risky. The LLM may use general knowledge about social patterns, but that knowledge may include stereotypes or correlations that are not appropriate for individual prediction. This is why fairness analysis and ablations are central to the project.

14 Fairness expectations

It is ambiguous whether BoL models will be more or less fair than tabular baselines.

They could become more fair if the richer individual history helps the model rely less on crude demographic group membership. For example, if two people have similar school, family, and behavioral histories, a rich BoL representation might make race or sex less decisive.

They could become less fair if the LLM imports external social stereotypes or uses proxy information more strongly. Even if explicit race and sex are removed, variables such as neighborhood, household structure, school context, or maternal background may still proxy protected attributes.

Therefore the recommended experimental design includes:

1. BoL with protected attributes included;
2. BoL with explicit protected attributes removed;
3. tabular baseline with protected attributes included;
4. tabular baseline with protected attributes removed;
5. fairness metrics for each condition.

15 Relation to LLM fine-tuning

When team members mention LLM fine-tuning, they most likely mean adapting an LLM to the specific task format using examples of Book-of-Life text and known outcomes. Fine-tuning could help the model:

- produce better-calibrated probabilities;
- follow the JSON output format more reliably;
- learn the specific meaning of NLSY variable descriptions;
- avoid overreacting to noisy or sparse information;
- make predictions more consistently across cases.

However, fine-tuning requires enough labeled examples and a clean train/test split. It also creates a risk of overfitting if the dataset is small. For the seminar project, a practical order is:

1. direct prompting on a pilot sample;
2. embeddings plus a simple classifier;
3. tabular baselines;
4. optional fine-tuning only if the simpler approaches are stable and the budget allows it.

16 Recommended baseline comparison

The next important step is to compare the current BoL pilot to tabular models under the same target definition and cutoff logic. A fair comparison should:

- use the same eligible children;
- use the same train/validation/test split;
- respect the same person-specific cutoff;
- use comparable predictor information;
- report AUC, accuracy, FPR, FNR, TPR, PPV, calibration, and fairness gaps.

Recommended tabular baselines are:

1. majority/base-rate classifier;
2. logistic regression with a limited theory-based feature set;
3. LASSO or elastic-net logistic regression;
4. gradient boosting or random forest;
5. possibly a model using BoL embeddings instead of raw text prompting.

The most scientifically useful comparison is probably not direct prompting versus tabular alone, but:

tabular model *vs.* text embedding + classifier *vs.* direct LLM prompting.

17 Limitations

17.1 Measurement limitations

The target combines self-reported behavior and justice-system contact. These are related but not identical. Justice contact depends not only on behavior but also on enforcement, detection, policing, charging, and sentencing. Therefore, models predicting justice contact may partially predict institutional exposure rather than behavior alone.

17.2 Missingness and survey routing

Missing values in NLSY data can mean many things: valid skip, non-interview, refusal, don't know, or structural ineligibility. Missingness is therefore not just noise. It may carry information about age, module eligibility, interview participation, or household situation. The current BoL texts mostly omit missing values rather than deeply modeling missingness. This is simple but may lose information.

17.3 Temporal leakage risk

The person-specific cutoff reduces leakage, but leakage remains a risk if a variable is coded with a survey year that does not perfectly correspond to the period it asks about. For example, an item asked in 2004 may ask about behavior since the last interview. Such windows need to be interpreted carefully.

17.4 Target-label balance

The 120-case pilot sample was balanced for experimental convenience. This means accuracy in the pilot is not directly equal to accuracy expected in the natural population distribution. For population performance, evaluation must be run on an unbalanced held-out sample with the natural base rate.

17.5 Threshold selection

The best threshold of 0.38 was selected on the same 120 cases used for evaluation. This is optimistic. A proper workflow should choose the threshold on validation data and report performance once on held-out test data.

17.6 Small subgroup fairness estimates

Fairness estimates are unstable in small groups. In the 120-case pilot, the Hispanic subgroup contains only 13 cases. Error-rate estimates for this group should be treated as descriptive pilot diagnostics, not final evidence.

17.7 LLM bias

The LLM may use patterns learned from external text that reflect social stereotypes. This is especially concerning for crime prediction. Removing explicit race and sex is not enough, because proxy variables remain. Therefore, protected-attribute ablations and subgroup error analysis are necessary.

18 Current file map

The most important current files are:

- `create_child_bol_broad_persistent.py`: creates the current broad-persistent BoL texts with person-specific cutoffs.
- `predict_child_bol_broad_persistent_openai.py`: runs direct LLM predictions on the 120-case sample.

- `evaluate_child_bol_broad_persistent_thresholds.py`: evaluates probability thresholds.
- `evaluate_child_bol_reoffense_from_original_120.py`: derives the reoffense labels from the original 120 prediction rows and evaluates thresholds for the reoffense cohort.
- `evaluate_child_bol_broad_persistent_fairness.R`: computes subgroup metrics and fairness gaps.
- `child_crime_broad_persistent_feature_index.csv`: lists the 864 BoL feature variables.
- `child_crime_broad_persistent_targets.csv`: stores target labels and cutoff years.
- `books_child_crime_broad_persistent_cutoff.json`: full generated BoL corpus for all eligible children.
- `books_child_crime_broad_persistent_sample_120.json`: balanced 120-case pilot sample.
- `child_crime_reoffense_llm_predictions_120.csv`: derived Option B predictions for the 85 first-event cases in the original 120-run file.
- `child_crime_reoffense_llm_threshold_eval_120.csv`: threshold evaluation for the derived reoffense target.
- `child_crime_broad_persistent_llm_predictions_120_with_reoffense.csv`: combined file containing the original Option A rows and the derived Option B rows.

19 Recommended next steps

1. Freeze the target definition for the next project milestone: keep `later_persistent_delinquency_contact` as Option A and report `later_reoffend_after_first_delinquency_contact` as Option B.
2. Create a proper train/validation/test split at the child level.
3. Rebuild BoLs for all eligible children with the same cutoff logic.
4. Build a dedicated reoffense cohort and run fresh Option B prompts, rather than relying only on the derived relabeling of the original 120 runs.
5. Run a tabular logistic/LASSO/boosting baseline using the same eligible sample and the same pre-cutoff information.
6. Compare year-specific features to collapsed question-level features in the tabular baselines.
7. Run the LLM or embedding model on a larger held-out test set.
8. Compare performance and fairness under four conditions: protected attributes included/removed for both tabular and BoL models.
9. Add calibration metrics, especially Brier score and calibration curves.
10. Consider a stricter predictor set excluding all prior delinquency/contact variables to answer a more purely preventive question.

20 Short interpretation

The current workflow shows that the Book-of-Life approach is feasible for this project. The strongest improvement came from making the data longitudinally correct and substantively richer: using person-specific cutoffs, expanding the feature set to relevant life-course domains, grouping the text by year and domain, and calibrating the LLM prompt. On a balanced 120-case pilot for persistent later delinquency/contact, the revised direct-prompting model achieved AUC 0.895 and best-threshold accuracy 0.858.

The current direction is to present this as the main Option A result and add the reoffense-after-first-event framing as Option B. The derived 120-run reoffense analysis gives a first indication that the same probabilities carry signal for reoffense, but it should be followed by a dedicated reoffense prompt and held-out evaluation. We also now distinguish between the 864 year-specific feature columns and the 135 collapsed question-level features, which makes the feature-design discussion clearer.

These results are promising, but they should be presented as pilot evidence. The main scientific contribution will come from a careful comparison against tabular baselines on held-out data, with leakage-aware cutoffs and fairness metrics reported transparently.

References

- [1] Capaldi, D. M., Tiberio, S. S., Kerr, D. C. R., & Owen, L. D. (2016). Understanding persistence and desistance in crime and risk behaviors in adulthood: Implications for theory and prevention. *Prevention Science*, 17, 785–793.
- [2] Pekala, K., et al. (2021). Risk factors of early adolescence in the criminal career of Polish offenders in the light of life course theory. *International Journal of Environmental Research and Public Health*, 18(12), 6583.
- [3] National Longitudinal Surveys. NLSY79 Children and Young Adults documentation, including question names and reference numbers.
- [4] Seminar starter materials. *Writing the Book of Life: Predicting life outcomes with LLMs and rich panel data*.